Project Milestone 01

Group Name: 2by2	Lab Section: EF - 6L
Member 1: Abucay, Mark Leo	Member 2: Labramonte, Nathalie Faith
Member 3: Reyes, Thomas III	Member 4: Solmoro, Kriztian Mer

Dataset Information:

- Title: BIR Tax Collection Data
- Source: Bureau of Internal Revenue (BIR), Philippines, Better Gov PH Open Data Portal
- Direct Link: https://data.bettergov.ph/datasets/20

Objectives:

- **Classification:** Predict if the regional tax collections in a given period will increase or decrease in comparison to the same month of the previous year, using past collected data
    - 2 Classes of the Target Variable:
        - Up: Increase of tax collection compared to the same month of the previous year
        - Down: Decrease or no change in the tax collection compared to the same month of the previous year
- **Clustering:** Group regions into distinct economic profiles based on their tax collections from 2020 - 2024 in categories such as mean monthly tax collection, average annual growth rate, and collection volatility (standard deviations of monthly collections relative to the mean). Profile each cluster after evaluating using the silhouette score and elbow method.

Methods of Algorithms:

Method	Suitability	Evaluation Metrics
Naive Bayes Classifier	• fast probabilistic method that is suitable for simple numeric and categorical features • can be used to model the independent probability distributions of each month/region in order to determine the likelihood of their revenue going "up" or "down"	• Accuracy: ○ $\frac{TP+TN}{TP+TN+FP+FN}$ ○ measures how often is each classification method correct • Precision: ○ $\frac{TP}{TP+FP}$ ○ examines how many of the predicted "up"
K-Nearest Neighbors (KNN)	• does not assume anything about the data	

	● can capture recurring seasonal patterns and regional similarities by matching current collection periods to similar past periods	cases were actually "up" ● Recall: ○ $\frac{TP}{TP+FN}$ ○ examines how many of the actual "up" data were successfully identified ● Macro-F1 Score: ○ $\frac{F1_{Up}+F1_{Down}}{2}$ ○ where F1 = $2\ x\ \frac{Precision\ x\ Recall}{Precision+Recall}$ ○ gives equal importance to "up" and "down" cases even when they have different number of observations ● Confusion Matrix: ○ 2 x 2 matrix in order to track True Positives, False Positives, True Negatives and False Negatives ○ identifies the type of mistakes made by each model
Perceptron	● learns a straight-line boundary between "up" and "down" by fixing its weights every time it makes a mistake ● acts as a linear baseline in order to test if revenue directional shifts are linearly separable	
K-Means Clustering	● can be useful for clustering regions with similar tax collection patterns into distinct economic profiles based on average monthly	● Silhouette Score ○ measures how well-separated the resulting clusters are ● Inertia (WCSS) with Elbow Method

	collections, growth rates, and volatility	○ identifies how many clusters should be used

Performance Comparison

Classification:

- All classifiers will use the same features
- All classifiers will use the same train/test split
  - Train: 2020-2023 Data
  - Test: 2024 Data
- Compare the models with the use of accuracy, precision, recall and F1-score
- Compare confusion matrices to determine how each model incorrectly predicts an increase or decrease
- Compare each model against a majority-class baseline to evaluate whether it performs better than naive guessing

Clustering:

- The regional features will be standardized before the application of K-Means in order for variables with larger numerical scales to not dominate the results
- Examine how inertia changes with different number of clusters using the elbow method
- Assess how well-separated the created clusters are using the silhouette score
- Analyze and compare the results based on their mean tax collection, growth rate and volatility

GitHub Repository:

- https://github.com/nthslies/cmsc170-project-bir-tax-collection.git